Saatwik Sairaam Vasamsetti

Full-Stack Web Developer · Software Engineer

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PROFILE

Full-stack engineer who ships production-shaped systems and measures them. Recent work spans Trade Research, where Python computes every figure and a verifier traces each one in a generated report back to its source data, and an offline-first emergency platform whose TypeScript/PostGIS API passes 636 assertions with a Kotlin Compose client and a YOLO11 pipeline beside it. Comfortable across the stack — API and schema design, transactional concurrency, front-end delivery, Python data pipelines and BI — and in the habit of reporting the result even when the result is negative.

EDUCATION

VIT-AP University, Amaravati *2024 – 2029*
Integrated M.Tech — Computer Science & Engineering (in collaboration with Virtusa) *CGPA: 7.55 / 10*
Coursework: *Advanced Programming, Data Structures & Algorithms, Database Management Systems, Software Engineering, Frontend Design & Testing*

SR Junior College — Intermediate (Class XII, MPC) *2022 – 2024 · 94.8%*
DAV Public School — Secondary School (Class X) *81.17%*

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Kotlin, SQL, Java, HTML, CSS
Backend & APIs: Node.js, Fastify, Express, FastAPI, REST design, JWT auth with rotating refresh tokens, RBAC, Zod validation, webhooks
Frontend: React 19, Next.js 16, Angular, Tailwind CSS, Jetpack Compose, progressive web apps, dependency-free canvas UI, accessibility (A11y)
Databases: PostgreSQL 16 + PostGIS, MySQL, Drizzle, schema & index design, transactional concurrency control
Data & ML: pandas, NumPy, scikit-learn, Ultralytics YOLO11, ONNX, exploratory data analysis, walk-forward back-testing, conformal prediction, Monte Carlo simulation
Visualisation & BI: Power BI, Tableau, Matplotlib, Chart.js, Advanced Excel dashboards, data storytelling
Cloud, Tooling & QA: AWS, Docker & Compose, Git, GitHub Actions CI, Gradle, Vitest, ESLint, PyInstaller

SELECTED PROJECTS

Trade Research — Verifiable Research & Execution Platform *Aug – Sep 2026*
Stack: *Python, scikit-learn, pandas, FastAPI, PostgreSQL, React, MetaTrader 5, PyInstaller*

- Built a leakage-free walk-forward back-test benchmarking six scikit-learn models against naive and drift baselines, with conformal prediction intervals reporting honest held-out coverage, Monte Carlo price paths, and a cost-aware trading simulation reporting Sharpe and drawdown.
- Inverted the usual "AI analyst" design so Python computes and the model only interprets: indicators are calculated in code, financials come from SEC EDGAR XBRL with accession numbers, missing data stays null, and a verifier machine-checks every figure in the finished note back to the snapshot.
- Implemented the execution platform end to end — FastAPI, PostgreSQL and React wiring alerts through risk checks and an OMS to the broker — plus an autonomous MetaTrader 5 demo harness with scheduled wind-down.
- Reported the negative result rather than burying it: none of the strategies measured here has a demonstrable edge, and the README says so first.
- Measured the packaging decision instead of preferring one: PyInstaller onedir starts in 1,000 ms against onefile's 5,400 ms, so the 24-command toolkit ships as a directory.

Private repository — available on request

RoadAssist Bharat — Offline-First Emergency Mobility Platform *Aug – Sep 2026*
Stack: *TypeScript, Fastify, PostgreSQL 16 + PostGIS, Drizzle, Kotlin, Jetpack Compose, Python, YOLO11, Docker, GitHub Actions*

- Designed and built the entire platform solo across 117 commits: a 64-route Fastify modular monolith, a 56-table PostGIS schema with 138 indexes and 5 migrations, four browser surfaces, a native Android client and a computer-vision pipeline.
- Verified the API with 636 assertions across six suites, zero failures — including 74 security attacks that must each be refused (cross-tenant access, role escalation, SQL injection, forged and alg:none tokens, unsigned webhooks) and 75 concurrency tests.
- Settled every race in the database rather than by timing: SELECT ... FOR UPDATE on booking rows with offer expiry re-checked under the lock, a partial unique index making "paid once" true under simultaneous confirmations, and a hash-chained audit log that Postgres RULES make immutable.
- Trained four YOLO11 variants on RDD2022 across four countries: the best reaches a measured mAP50 of 0.471 (mAP50-95 0.226) on held-out validation at ~135 ms/image, and a separate nano tier trades accuracy for ~48 ms/image on CPU for Pi-class hardware. Potholes remain the hard class at ~0.24, and the per-class breakdown says so.

- Shipped Off-Grid Mode as architecture, not fallback: a Kotlin SOS ladder (data → SMS → 112 → offline queue) where exactly one channel owns an emergency, proven by 21 unit tests on that file alone, so a retry never dispatches a second ambulance.
- Wrote CI gates that fail the build when a document disagrees with the measured figures, when a file:line citation rots, or when module boundaries break — after the same wrong number reached twenty documents three times and a human caught it each time.
- Made a feature phone a first-class client: a complete booking over SMS with no app, in eight languages switched by text command — seven of them machine-translated and not yet native-reviewed, which the project states rather than implies.

github.com/saatwik-1157/roadassist-bharat

MachineSense AI — Explainable Industrial Intelligence *Smart India Hackathon 2026*

Stack: *Python, FastAPI, JavaScript, Canvas*

- Built an Industry 4.0 prototype for Indian industrial units: a FastAPI backend simulating a fleet of IoT-instrumented machines (vibration, temperature, current, sound, power) behind an analytics engine and a six-view real-time dashboard.
- Implemented the AI layer — a weighted 0-100 machine health score, MAD z-score anomaly detection, remaining-useful-life and failure-probability forecasting, and energy/cost/CO2 optimisation across the fleet.
- Made every score explainable: each health number is decomposed by sensor with plain-language root-cause reasoning, and a digital twin simulates a maintenance action before anyone touches the real machine.
- Mirrored the same scoring logic in dependency-free browser JavaScript so the demo behaves identically with the backend down, and labelled on screen which mode is running.

github.com/saatwik-1157/machinesense-ai

Smart University Management System (SUMS) *2026*

Stack: *MySQL, JavaScript, Chart.js, CSS3*

- Designed a normalised MySQL schema for an enterprise university system managing students, courses, faculty and academic records, with the query layer behind it.
- Built a glassmorphism front end with Chart.js dashboards surfacing enrolment and performance insights over the live data.

github.com/saatwik-1157/database_management_SUMS

OfferPilot — Job-Application Platform for F-1 / OPT Students *2026*

Stack: *JavaScript, HTML, CSS, localStorage*

- Built a dependency-free static platform — marketing site, onboarding, client dashboard, profile, legal pages and a passcode-gated ops console — with no build step and full localStorage persistence.
- Shipped the RezForge resume-tailoring engine, an interview kanban, a referral programme and analytics charts on a hand-written SVG engine, plus an OPT Runway Calculator computing work-authorisation runway, unemployment-day budget and H-1B lottery timing.
- Labelled the prototype honestly on the page: the match and resume engines are deterministic mocks and no real applications are submitted.

github.com/saatwik-1157/offerpilot

Angular Applications — ShopCart, Notes, Search & Auth *2026*

Stack: *Angular 22, TypeScript 6, standalone components, signals, Vitest*

- **ShopCart (e-commerce):** searchable, category-filtered catalogue, product detail pages, a quantity-aware cart with computed totals, and signup/sign-in with a route guard protecting checkout — 14 Vitest unit tests across cart, accounts and header.
- **Search template:** a user list loaded from JSON through a service and filtered on screen as the query is typed, matching on either name or email address.
- **Notes app:** add, update and delete notes held in a service-owned signal, rendered through separate form, list and single-note components.
- **Authentication app:** a login page validated in a service, with the home page reachable only through a route guard after a successful sign-in.
- Every experiment's lab record is generated from the live source on disk rather than written by hand, so the submitted document cannot disagree with the code that ran.

ExamPlanner — Android Study Planner *2026*

Stack: *Kotlin, Android, Java*

- Built a native Android study planner with authentication, exam and schedule management, and a Pomodoro timer with customisable focus and break intervals.

GenAI-Powered Data Analytics — Job Simulation *Forage · Jun 2026*

Stack: *Python, pandas, GenAI, data storytelling*

- Performed exploratory data analysis and risk profiling on a real-world lending dataset, built a model to predict customer delinquency, and delivered the data storytelling behind an AI-driven collections strategy.

Additional projects

- PathAhead — production-ready responsive site for a career-guidance platform running workshops for intermediate students across India (41 commits).
- Gemini Chat Client — a complete browser-only chat client: multi-thread sidebar, four personas, vision and voice input, Markdown rendering sanitised with DOMPurify, localStorage persistence, no server and no build step.

EXPERIENCE & LEADERSHIP

Campus Ambassador — SmartED *Jun 2026*

- Led campus outreach for SmartED's initiatives; formally recognised for leadership, dedication and the connections built across the student community.

Data Analytics Job Simulations — Deloitte, Quantium & GenAI programmes via Forage *Jun – Aug 2026*

- Deloitte (Jul 2026): completed practical tasks in data analysis and forensic technology.
- Quantium (Aug 2026): data preparation and customer analytics, experimentation and uplift testing, and the commercial application of the results.
- GenAI-Powered Data Analytics (Jun 2026): exploratory data analysis and risk profiling on a lending dataset, delinquency prediction, and the data storytelling behind an AI-driven collections strategy.

CERTIFICATIONS

• Google Analytics Certification — Google, May 2026 – May 2027 (ID 183723043)
• GenAI-Powered Data Analytics (Job Simulation) — Forage, Jun 2026
• Data Analytics Job Simulation — Deloitte via Forage, Jul 2026
• Data Analytics Job Simulation — Quantium via Forage, Aug 2026
• Introduction to AWS — AWS Training & Certification, May 2026
• AWS Certified Cloud Practitioner (CLF-C02) Domain 1 Review — AWS, May 2026
• AWS Certified Data Engineer – Associate (DEA-C01) Exam Prep — AWS, May 2026
• Exploring SAP Analytics Cloud — SAP Learning, Jun 2026
• Becoming an SAP Data Architect — SAP Learning, Jun 2026

ACHIEVEMENTS

- Sole author of RoadAssist Bharat — 117 commits covering API, Android client, CV pipeline, schema, six test suites and CI, with 636 assertions passing over the API and zero failures.
- Trained and measured four YOLO11 detector variants on RDD2022, reaching mAP50 0.471 on held-out validation and publishing the per-class breakdown including the classes that perform badly.
- Earned nine professional certifications across data analytics and cloud — AWS, Google and SAP — including three Forage job simulations (Deloitte, Quantium, GenAI analytics).
- Participant, HackDevengers 1.0 hackathon — organised by Devengers, 2026.
- Scored 94.8% in Class XII and 81.17% in Class X; recognised as Campus Ambassador, SmartED (2026).
- Designed, built and deployed a dependency-free personal portfolio at saatwik-1157.github.io.